

EXPERIENCE

- **Genbio AI** Paris, France
Research Scientist *Sep 2025 – Now*
 - **Multi-Modal Foundational Models:** Developed and trained large-scale (billions of parameters and data points) multi-modal foundational models for biological applications.
- **Meta AI** California, USA
Research Scientist Intern *Aug 2023 – Feb 2024*
 - **3D and Generative Learning:** Developed methods to integrate self-supervised learning and 3D generative modeling, focusing on mutual benefits for reconstruction and synthesis tasks.
- **Institute for Adaptive and Neural Computation** Edinburgh, UK
Postgraduate Researcher *Sep 2021 – Present*
 - **Shape and Semantics in ML:** Developed models for articulated 3D object reconstruction and explored the role of shape in semantic tasks, including fine-grained classification and correspondence.
- **TUM Computer Vision Lab** Munich, Germany
Research Assistant *Oct 2018 – Jun 2021*
 - **Holistic Scene Understanding:** Worked on Lidar-based semantic segmentation, instance segmentation, and multi-object tracking.

SELECTED PUBLICATIONS

- D. Duolikun, **M. Aygun**, C. Li, H. Bilen, and O. Mac Aodha. “DepthCues: Evaluating Monocular Depth Perception in Large Vision Models.” **CVPR 25’**
- **M. Aygun**, O. Mac Aodha, “SAOR: Single-view Articulated Object Reconstruction”, **CVPR 24’**
- **M. Aygun**, O. Mac Aodha, “Demystifying Unsupervised Semantic Correspondence Estimation”, **ECCV 22’**
- **M. Aygun**, A. Osep, M. Weber, M. Maximov, C. Stachniss, J. Behley, L. Lael-Taixe, “4D Panoptic Lidar Segmentation”, **CVPR 21’**
- **M. Aygun**, Y. Aytar, H. K. Ekenel, “Exploiting Convolution Filter Patterns for Transfer Learning”, **ICCV 17’ Workshop on Transfer Learning - (Honourable Mention Award)**

EDUCATION

- **The University of Edinburgh** Edinburgh, UK
PhD in Computer Science *Sep 2021 – June 2025*
- **Technical University of Munich** Munich, Germany
MSc in Computer Science *Oct 2018 – Dec 2020*
- **Istanbul Technical University** Istanbul, Turkey
BSc in Computer Science *Sep 2013 – Jun 2017*

SKILLS

- Technical: Python, PyTorch (3D), C/C++, OpenCV, NumPy, Matlab, Linux, Git
- Methodologies: Generative modeling, self-supervised learning, 3D reconstruction, shape analysis

AWARDS & ACTIVITIES

- Organizer for the 12th Fine-Grained Visual Categorization Workshop at CVPR 2025
- Reviewer for TPAMI, CVPR, ICCV, ECCV
- Tutoring for Applied Machine Learning Course at the University of Edinburgh. 2021-2024